GEOMETRY · CIRCLES

# Circle Angle Explorer

Explore & Reflect

Name _____ Date _____ Period _____

Open **Circle Angle Explorer** on Mr. K's Classroom: Geometry → Circles → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

## 1 Predict before you touch the tool

1. A central angle and an inscribed angle intercept the same 100° arc. Which is bigger, and by how much?

## 2 Explore use the tool and record what you see

### Central and inscribed angles

Drag A and B to set each arc.

| Arc AB | Central ∠AOB | Inscribed ∠APB |
|---|---|---|
| 120° | | |
| 100° | | |
| 60° | | |
| 180° (AB is a diameter) | | |

### Arc length and sector area

Use the second tab. Leave answers in terms of π.

| r | Angle | Fraction of circle | Arc length | Sector area |
|---|---|---|---|---|
| 4 | 90° | | | |
| 6 | 60° | | | |
| 3 | 120° | | | |
| 5 | 180° | | | |

## 3 Write the rule fill in each blank from what you observed

1. An inscribed angle is _____ the measure of its intercepted arc.
2. A central angle is _____ its intercepted arc.
3. An inscribed angle that intercepts a semicircle measures _____.
4. Inscribed angles that intercept the same arc are _____.
5. Arc length = $\frac{\text{angle}}{360°}$ · _____ and sector area = $\frac{\text{angle}}{360°}$ · _____.

## 4 Check no tool: show your work

1. An inscribed angle measures 35°. Find its intercepted arc.

2. Find the arc length when r = 9 and the central angle is 40°.

3. Find the area of a sector with r = 10 and a 72° central angle.

4. Triangle ABC is inscribed with AB a diameter and ∠A = 25°. Find ∠B.

## 5 Reflect

Why doesn't the inscribed angle change when P slides along the circle?